Deferential Equation (DE): A differential equation is an equation which involves derivatives.

Ex:

1. $\frac{dy}{dx} = 0$
2. $\frac{dy}{dx} + y = x$
3. $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$
4. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$

Classification of DE: The classification of DE is based ~~on~~ on the natual of the derivatives in the equation. There are two classes of DE:

Ordinary differential equation (ODE):

1. $\frac{dy}{dx} = 0$
2. $\frac{dy}{dx} + xy = \cos x$
3. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 5$

Partial differential equation (PDE):

1. $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$
2. $\frac{\partial z}{\partial t} = c\frac{\partial^2 z}{\partial x^2}$
3. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$

Order of the DE: The order of DE is the order of the maximum derivative involved.

1. $\frac{dy}{dx} = 0 \longrightarrow$ 1st order
2. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0 \longrightarrow$ 2nd order

Degree of DE: The degree of a DE is the power of the maximum derivative.

① $\left(\frac{dy}{dx}\right)^1 = 0 \longrightarrow$ 1st degree ODE

② $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^5 + y = \sin x \longrightarrow$ 3rd degree ODE

Formation of ODE: An ordinary differential equation (ODE) is formed by differentiating the equation and eliminating arbitrary constants.

Ex ① Form an ODE of $y = c$.

Ans: Given $y = c$ —— ①

Differentiating ①, w.r.t. x

$\boxed{\frac{dy}{dx} = 0}$ This is a first order and 1st degree ODE.

Ex ② Form an ODE of $y = A\cos x + B\sin x$

Ans) Given equation is $y = A\cos x + B\sin x$ —— ①

Eq^n ① is an explicit function with two arbitrary constants, so from ①, we will get a 2nd order ODE.

Diff. ① w.r.t. x,

$$\frac{dy}{dx} = -A\sin x + B\cos x$$

Again, diff.

$$\frac{d^2y}{dx^2} = -A\cos x - B\sin x$$

$$= -(A\cos x + B\sin x)$$

$$\frac{d^2y}{dx^2} = -y \quad [\text{using } 1]$$

$\Rightarrow \boxed{\frac{d^2y}{dx^2} + y = 0}$ which is a 2nd order and 1st degree ODE.

<u>H.T.</u> Form the ODE for:

1. $y = e^x(A\cos x + B\sin x)$
2. $y = e^{\frac{x}{a}} - e^{-\frac{x}{a}}$

<u>Solution of 1st order ODE:</u> The solution of ODE is the relation between the dependent and independent variables which is free from derivatives. There are different types of solution procedure of the 1st order and 1st degree ODE as follows:

1. Variable seperable method
2. Homogeneous ODE
3. Linear ODE
4. Exact ODE

For solving ODE, we have to remove the deriative by applying the suitable integration. There are some ~~rules~~ formulas for integration as follows:

1. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c$

2. $\int \frac{dx}{\sqrt{a^2+x^2}} = \ln\left|x+\sqrt{a^2+x^2}\right| + c$

3. $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left|x+\sqrt{x^2-a^2}\right| + c$

4. $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$

5. $\int \sqrt{a^2+x^2}\,dx = \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\ln\left|x+\sqrt{a^2+x^2}\right| + c$

6. $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + c$

7. $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + c$

8. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$

9. $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + c$

10. $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$

variable seperable method:

① $\frac{dy}{dx} = f(x)\, g(y)$

$\Rightarrow \boxed{\frac{dy}{g(y)} = f(x)\, dx}$

$f(x) * g(y)$

$g(y)\, dx + f(x)\, dy = 0$

$\Rightarrow \boxed{\frac{dx}{f(x)} + \frac{dy}{g(y)} = 0}$

Applying integration

Ex: ① $\overset{xy}{y\, dx + x\, dy} = 0$

$\Rightarrow \frac{dx}{x} + \frac{dy}{y} = 0$

Integrating, we get

$\int \frac{dx}{x} + \int \frac{dy}{y} = 0$

$\Rightarrow \ln x + \ln y = \ln C.$

$\Rightarrow \ln(xy) = \ln C$

$\therefore \boxed{xy = c}$

Ex 2: $(1+y^2)\, dx + \sqrt{1-x^2}\, dy = 0$

$\Rightarrow \frac{dx}{\sqrt{1-x^2}} + \frac{dy}{1+y^2} = 0$

Integrating,

$\int \frac{dx}{\sqrt{1-x^2}} + \int \frac{dy}{1+y^2} = 0$

$\Rightarrow \sin^{-1}(x) + \tan^{-1} y = C.$

Ex 3: $(1+x)\,dy - y\,dx = 0$

$$\Rightarrow \frac{dy}{y} - \frac{dx}{1+x} = 0$$

Integrating,

$$\ln y - \ln(1+x) = \ln c$$

$$\Rightarrow \ln\left(\frac{y}{1+x}\right) = \ln c$$

$$\Rightarrow \frac{y}{1+x} = c$$

$$\Rightarrow y = c(x+1)$$ Ans

Initial value problem (IVP): A differential eqn with initial condition is known as IVP.

Ex: $\left.\begin{array}{l} \dfrac{dy}{dx} = f(x)g(y) \longrightarrow \text{ODE} \\ y(x_0) = y_0 \longrightarrow \text{I.C.} \end{array}\right\} \longrightarrow \text{IVP}$.

Ex: Solve $\dfrac{dy}{dx} = -\dfrac{x}{y}$; $y(4) = 3$

Ans: $\dfrac{dy}{dx} = -\dfrac{x}{y}$

$$\Rightarrow y\,dy = -x\,dx$$

Integrating, $\dfrac{y^2}{2} = -\dfrac{x^2}{2} + c$.

at $x = 4$, $y = 3$, $\dfrac{3^2}{2} = -\dfrac{16}{2} + c$

$$\Rightarrow c = 25/2$$

$$\therefore \quad \frac{y^2}{2} = -\frac{x^2}{2} + \frac{25}{2}$$

$$\Rightarrow \boxed{y^2 + x^2 = 25}$$

H.T.

1. $e^{-y}\,dx + (1+x^2)\,dy = 0$
2. $\dfrac{dy}{dx} = x^2(e^{3y} - 1)$
3. $x \cos^2 y \, dx = y \, dy$
4. $\dfrac{dy}{dx} = \dfrac{y^2 + y + 1}{x^2 + 2x + 1}$
5. $\dfrac{dy}{dx} = -4xy^2 \; ; \; y(0) = -1.$